sci

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

Materials that are both living and non living, exploring the possibility of an extraterrestrial impactor at the dawn of civilisation, how organisms sync with the Sun and hunting for Martians

Skittering water on oily pans

The commonplace phenomenon of droplets of water in a hot oily pan sizzling and sliding rapidly across the surface has been quantified for the first time. The deeper understanding of the phenomenon could be used to improve micro-fluidic systems and heat transfer devices.
https://dgit.in/h2oil

WHAT'S NEW

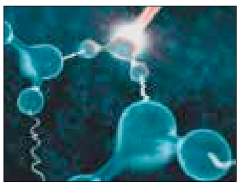
Researchers develop "neurograins" brain computer interface

Researchers from Brown university have developed and tested a new kind of brain computer interface on mice. Conventional BCI systems use one or two implantable sensors that monitor a few hundred neurons in a few selected locations of the brain, which can then be used to control prosthetic limbs or communicate with computers. The researchers have developed a new system that uses a cluster of independent implants, each of which are the size of a grain of salt. The network of microscale neural sensors independently record neural activity and transfer the data

wirelessly to a central hub that coordinates and processes the signals. These new types of sensors have miniaturised circuitry for detecting, amplifying and transmitting the brain signals on salt grain sized silicon chips, each with a unique network address. The hub is a thin patch about the size of half stuck externally on the scalp.
https://dgit.in/neurograins

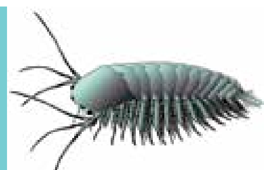
Scientists probe quantum tug between water molecules

Water is the most abundant substance on the planet, as well as one of the least understood. Many of the strange properties of water are derived from a hydrogen bonding between oxygen atoms in one molecule and hydrogen atoms in a neighbouring molecule, apart from the covalent bonds within the molecule. A team of international researchers have probed the hydrogen bonding between water molecules for the first time, with an effort to better understand the strange properties of water. When excited with laser light, the hydrogen atoms within water push and pull at the neighbouring molecules, providing a window into the microscopic origins of water's strange properties. Although the nuclear quantum effect was believed to be the root cause of water's strange properties, this is the first time that scientists have directly observed it at work.
https://dgit.in/wtrprb



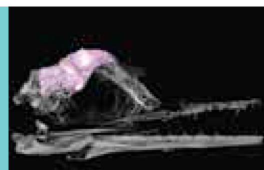
Ornithischians breathed different

Researchers have found the fossil remains of an early ancestor of triceratops and stegosaurs with a unique breathing mechanism. This turkey sized dinosaur had what are known as "pelvic bellows", with both the abdomen and the lungs expanding and contracting.
https://dgit.in/pbd



Ancestral arthropod brains

Researchers have found rare fossilised brain tissue of an ancestral arthropod, from which insects, crustaceans and millipedes have all descended, and discovered a discrete prosocerebral brain region, which is only seen in the embryonic development of modern critters.
https://dgit.in/arthbrn



Birds had advanced brains

Despite the phrase "bird brained", birds actually have by far the most advanced brains in the animal kingdom apart from mammals. In fact, researchers now believe that it was because of their brains that birds escaped the extinction that killed the dinos.
https://dgit.in/brdbrn